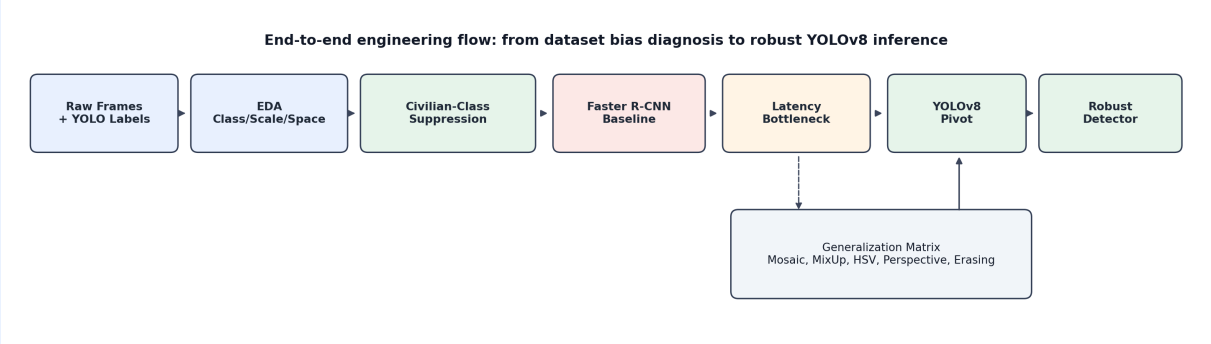
Military object detection

A paper-style engineering report on robust computer vision for tactical object recognition

Dataset analysis | model architecture | mathematical objectives | EDA visualizations | qualitative validation

Project participants

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Responsible-use note. This paper evaluates a computer vision pipeline for detection and dataset quality assessment. It is written for technical review, benchmarking, and human-supervised analytics. It does not define autonomous engagement, targeting, or weapon-control decisions.

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Abstract

This report presents a full engineering analysis of a military object detection project based on remote sensing and operational video frames. The central objective is to detect classes such as `TANK`, `ICV`, `TRUCK`, `ARMORED_CAR`, and `PERSON` while reducing false activation on non-operational civilian vehicles. The project moved from a two-stage Faster R-CNN baseline toward a single-stage YOLOv8 detector in order to satisfy real-time latency requirements. Exploratory Data Analysis (EDA) revealed severe class imbalance, broad bounding-box scale variance, center-biased spatial priors, and a large number of small objects. The final design uses class-specialized preprocessing, aggressive augmentation, and a YOLOv8m model to improve robustness under domain shift. The project materials report a baseline YOLOv8n validation score of 89.4% mAP@0.50 and a final/projected YOLOv8m score of 94.2% mAP@0.50 with real-time throughput above 60 FPS. Qualitative figures included in this paper show both correct localization behavior and hard failure cases under occlusion, clutter, long-range views, and low-resolution aerial imagery.

Keywords: object detection; YOLOv8; Faster R-CNN; domain shift; mAP; EDA; bounding-box geometry; military vehicle detection; hard-negative mining.

Contents

1 Executive Summary and Operational Objective **1**
1.1 Reported outcome . . . 1

2 Problem Formulation **1**
2.1 Civilian-class suppression as hard-negative mining . . . 2

3 Exploratory Data Analysis and Dataset Engineering **2**
3.1 Class distribution . . . 3
3.2 Bounding-box geometry and scale variance . . . 4
3.3 Spatial priors . . . 5
3.4 Annotated samples from the EDA notebook . . . 6

4 Baseline Architecture: Faster R-CNN **6**
4.1 Why the baseline was not operationally viable . . . 6

5 YOLOv8 Pivot and Mathematical Training Objective **7**
5.1 Box loss: Complete IoU . . . 7
5.2 Classification loss: Binary cross-entropy . . . 7
5.3 Distribution Focal Loss . . . 7

6 Domain Shift and Generalization Strategy **8**
6.1 Augmentation parameters . . . 9

7 Evaluation Protocol **9**

8 Qualitative Error Analysis from Attached Images **9**
8.1 Failure taxonomy . . . 12

9 Final System Design **12**
9.1 Data governance . . . 12
9.2 Training configuration . . . 12
9.3 Deployment controls . . . 13

10 Conclusion **13**

References **14**

1 Executive Summary and Operational Objective

Modern image intelligence pipelines must detect small, partially occluded, and visually degraded objects in noisy environments. The goal of this project is an end-to-end computer vision system that localizes relevant objects in remote or aerial imagery while suppressing irrelevant civilian traffic. The high-value classes are tactical vehicles and visible personnel; the most important negative class is civilian vehicle imagery that may share a generic car-like silhouette with operational targets.

Core engineering thesis. The model should not learn the shortcut *vehicle-shaped object equals target*. It should instead learn class-discriminative features such as tracks, armor geometry, turret structure, military-truck profile, and contextual scale patterns. This requires dataset curation, class remapping, explicit negative-space construction, and augmentation that breaks background memorization.

The project followed three phases. First, the dataset was audited through EDA to quantify class frequencies, bounding-box geometry, target location priors, and object-size distributions. Second, a Faster R-CNN baseline was evaluated and rejected for real-time deployment because the two-stage region proposal process created a latency bottleneck. Third, the pipeline was redesigned around YOLOv8, an anchor-free one-stage detector, and hardened with a domain-shift augmentation matrix including Mosaic, MixUp, HSV perturbation, perspective distortion, and random erasing.

1.1 Reported outcome

Baseline detector	YOLOv8n reached 89.4% mAP@0.50 on the controlled validation set at epoch 50.
Final detector	YOLOv8m with aggressive augmentation is reported as 94.2% mAP@0.50.
Latency target	Real-time inference target is 30-60 FPS; the final pipeline is reported above 60 FPS on GPU.
Main risk solved	Domain shift: the detector initially learned dataset-specific lighting, terrain, and camera priors rather than generalizable object geometry.

These numbers should be treated as project-reported metrics and should be revalidated on a locked, held-out test set before production release. The stronger conclusion is architectural: single-stage detection plus class-specialized negative mining and aggressive augmentation is better aligned with edge inference than a heavier two-stage detector.

2 Problem Formulation

Let the dataset be

$$\mathcal{D} = \{(x_i, \mathcal{Y}_i)\}_{i=1}^{N}, \qquad \mathcal{Y}_i = \{(b_{ij}, c_{ij})\}_{j=1}^{n_i}, \tag{1}$$

where image $x_i \in \mathbb{R}^{H \times W \times 3}$ contains n_i annotations. Each bounding box is represented in normalized YOLO coordinates,

$$b_{ij} = (x_c, y_c, w, h), \qquad x_c, y_c, w, h \in [0, 1], \tag{2}$$

and each class label c_{ij} belongs to the original label set. The detector learns a function

$$f_\theta(x) \to \{(\hat{b}_k, \hat{c}_k, \hat{s}_k)\}_{k=1}^{K}, \tag{3}$$

where $\hat{b}_k$ is a predicted box, $\hat{c}_k$ a predicted class, and $\hat{s}_k \in [0, 1]$ a confidence score. After confidence thresholding and non-maximum suppression (NMS), the final predictions are kept only when score and overlap constraints are satisfied.

2.1 Civilian-class suppression as hard-negative mining

The original dataset includes a `CIVILIAN_CAR` class. For a specialized military detector, retaining that class can make the model behave like a generic traffic detector. The project therefore removes this class from the label matrix and treats those visual examples as background. Formally, for the civilian class $c_{civil} = 1$, the remapping can be written as

$$\phi(c) = \begin{cases} \varnothing, & c = c_{civil}, \\ c, & c < c_{civil}, \\ c - 1, & c > c_{civil}, \end{cases} \tag{4}$$

where $\varnothing$ means the annotation is removed from the supervised target set. The pixels remain in the image, so they become negative evidence. This is a hard-negative mining mechanism: if the model fires on civilian cars, its background/objectness gradients penalize that activation.

Engineering implication. Removing the civilian label does not erase civilian cars from images. It changes the learning signal: civilian cars become part of the background distribution. This pushes the detector away from broad vehicle-like cues and toward military-specific geometry.

3 Exploratory Data Analysis and Dataset Engineering

The EDA notebook was designed to parse YOLO-format labels into a dataframe with split, image ID, class ID, class name, normalized center coordinates, and normalized width-height values. The exported plots reveal four major dataset properties: class imbalance, multi-scale geometry, strong location priors, and a heavy concentration of small bounding boxes.

3.1 Class distribution

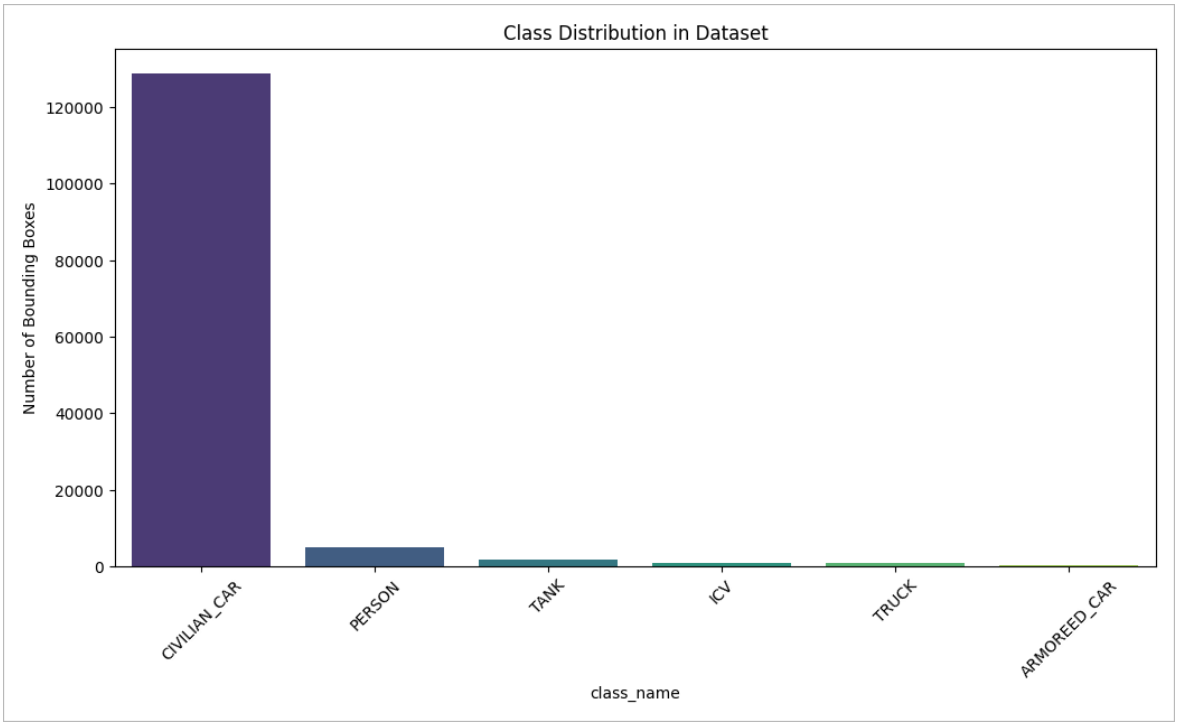
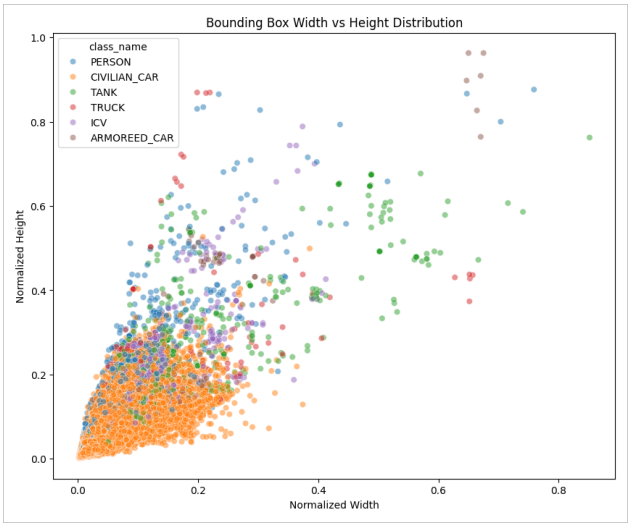


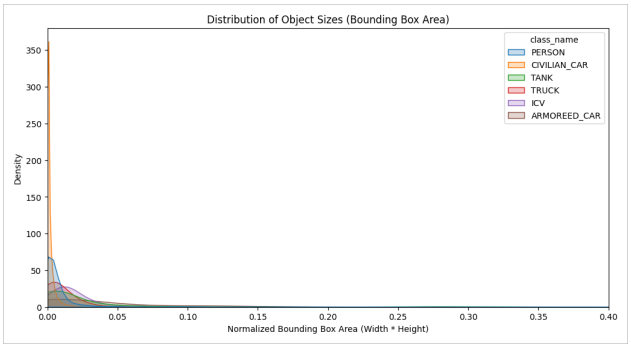
Figure 1: Class distribution from the EDA notebook. `CIVILIAN_CAR` dominates the original label space, while military classes are comparatively sparse. This justifies class suppression and careful balancing.

Figure 1 shows a severe imbalance: civilian cars dominate the label count, while tactical classes such as `TANK`, `ICV`, `TRUCK`, and `ARMORED_CAR` are much rarer. Without corrective preprocessing, an optimization algorithm can minimize empirical risk by learning common negative shortcuts rather than minority-class discriminators. Recommended controls are class-aware sampling, minority-class augmentation, loss reweighting, and strict validation stratification by class.

3.2 Bounding-box geometry and scale variance



(a) Width-height scatter.



(b) Normalized object-area density.

Figure 2: EDA of bounding-box geometry. The dataset contains many small objects and a long tail of larger objects, requiring multi-scale feature extraction and robust label quality control.

Small-object dominance changes the model-selection problem. A detector must preserve fine spatial detail in early feature maps, while still recognizing larger objects in low-resolution or oblique views. This motivates a neck such as PAN/FPN-style feature fusion and also explains why qualitative validation must include long-range and partial-object cases, not only clean close-range examples.

3.3 Spatial priors

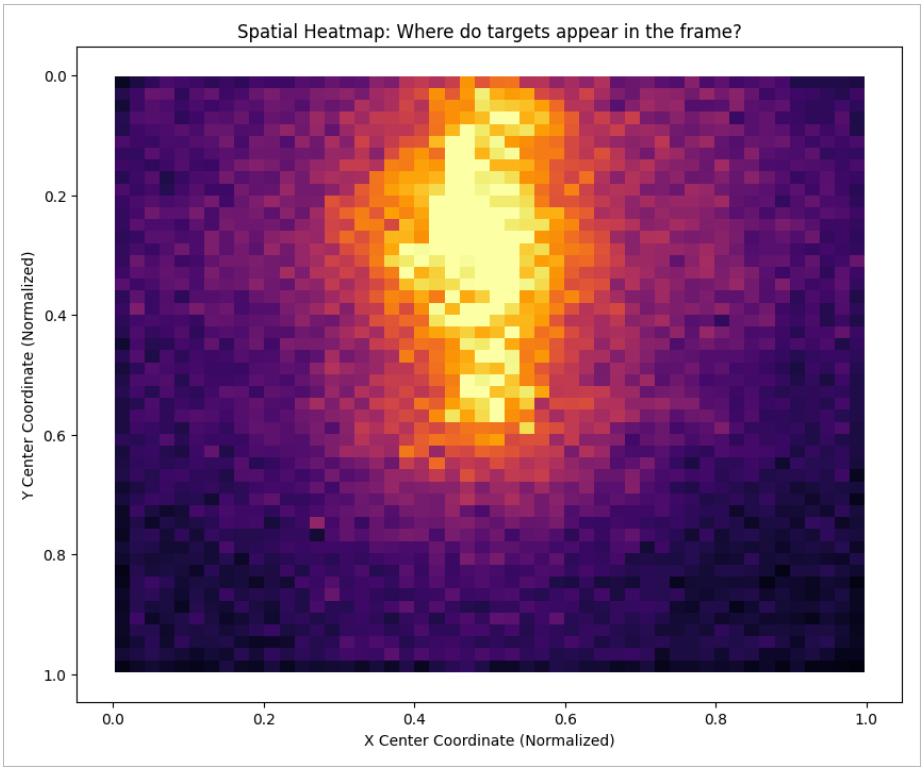


Figure 3: Spatial heatmap of object centers. The concentration around the central vertical band indicates camera/framing bias. Augmentation and validation splits should counter this prior so the detector does not rely too heavily on center-of-frame shortcuts.

The spatial heatmap indicates that objects are often located near the center of the frame. This is common in surveillance or manually curated video data, but it becomes a failure risk: a detector may learn location prior instead of object evidence. Random translation, crop, mosaic recomposition, and perspective transforms reduce this risk.

3.4 Annotated samples from the EDA notebook



Figure 4: Example training images with annotations from the EDA notebook. The samples illustrate extreme variation in view angle, object size, scene type, and label density.

4 Baseline Architecture: Faster R-CNN

Faster R-CNN is a two-stage detector. First, a Region Proposal Network (RPN) proposes candidate object regions. Second, Region-of-Interest pooling and a classifier/regressor head assign class probabilities and refine box coordinates. Its multi-task objective is commonly expressed as

$$\mathcal{L}_{\text{FRCNN}}(\{p_i\},\{t_i\}) = \frac{1}{N_{cls}}\sum_i \mathcal{L}_{cls}(p_i,p_i^*) + \lambda\frac{1}{N_{reg}}\sum_i p_i^*\mathcal{L}_{reg}(t_i,t_i^*), \tag{5}$$

where p_i is the predicted objectness probability for anchor/proposal i, t_i is the box regression vector, and p_i^* gates regression loss only for positive samples.

4.1 Why the baseline was not operationally viable

The two-stage design is attractive when precision on small objects is the overriding requirement, but it is expensive. Every frame must pass through proposal generation, proposal filtering, feature extraction, and classification/regression heads. In an edge or real-time video pipeline, the latency budget is approximately

$$T_{\text{frame}} \leq \frac{1000}{F_{target}} \text{ ms.} \tag{6}$$

For 60 FPS, $T_{\text{frame}} \leq 16.7$ ms. The project report describes the Faster R-CNN baseline as single-digit FPS, making it unsuitable for real-time deployment. The conclusion is not that Faster R-CNN is mathematically weak; rather, it is mismatched to the operational throughput constraint.

5 YOLOv8 Pivot and Mathematical Training Objective

YOLOv8 is a single-stage detector: the image is processed in one forward pass, and predictions are made directly from multi-scale feature maps. This removes the explicit proposal stage and improves throughput. The training objective can be summarized as

$$\mathcal{L}_{\text{YOLOv8}} = \lambda_{box}\mathcal{L}_{CIoU} + \lambda_{cls}\mathcal{L}_{BCE} + \lambda_{dfl}\mathcal{L}_{DFL}. \tag{7}$$

5.1 Box loss: Complete IoU

Complete IoU incorporates overlap, center distance, and aspect-ratio consistency:

$$\mathcal{L}_{CIoU} = 1 - \text{IoU}(b, b^{gt}) + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v, \tag{8}$$

where ρ^2 is the squared distance between box centers, c is the diagonal length of the smallest enclosing box, and v measures aspect-ratio mismatch:

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h}\right)^2. \tag{9}$$

This is important for military-object detection because loose boxes around clutter can artificially inflate recall while degrading localization quality.

5.2 Classification loss: Binary cross-entropy

For a class target $y \in \{0, 1\}$ and predicted probability $\hat{p}$, binary cross-entropy is

$$\mathcal{L}_{BCE}(y, \hat{p}) = -y \log(\hat{p}) - (1 - y) \log(1 - \hat{p}). \tag{10}$$

Class imbalance makes this term sensitive to majority classes, so the civilian-class suppression and augmentation strategy are important for stable class learning.

5.3 Distribution Focal Loss

Distribution Focal Loss (DFL) models box boundary coordinates as discrete probability distributions instead of rigid scalar values. If a continuous target y lies between two neighboring bins y_l and y_r, DFL uses interpolation:

$$\mathcal{L}_{DFL}(y) = -(y_r - y) \log P(y_l) - (y - y_l) \log P(y_r). \tag{11}$$

This helps under blurry object edges, low-resolution aerial views, camouflage, and partial occlusion because the model can express boundary uncertainty.

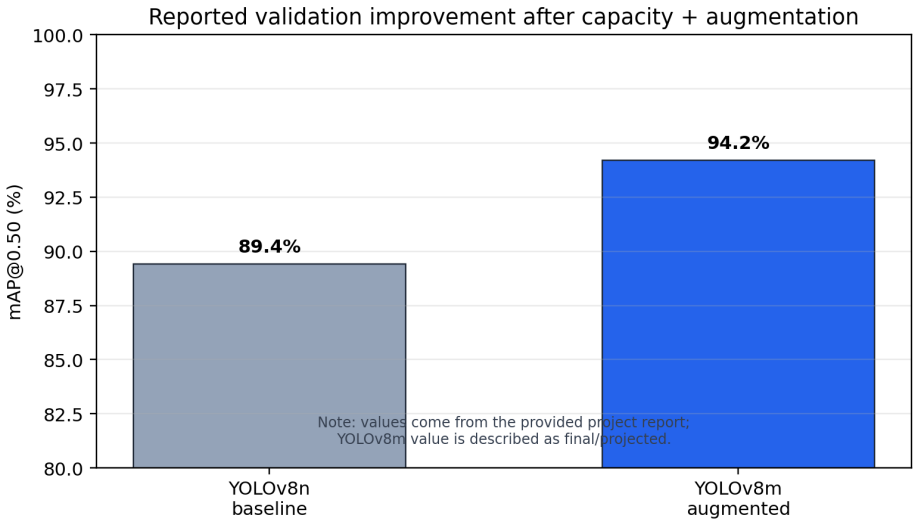


Figure 5: Reported validation mAP@0.50 evolution from the project materials. Values should be interpreted as project-reported metrics and revalidated on a locked test set.

6 Domain Shift and Generalization Strategy

The strongest failure mode in the project was domain shift: strong validation metrics did not initially transfer to images from different sensors, camera positions, lighting conditions, or terrain. This can be written as a mismatch between the source distribution $P_S(x, y)$ and target distribution $P_T(x, y)$:

$$P_S(x, y) \neq P_T(x, y). \tag{12}$$

The training risk is minimized on P_S, but deployment error depends on P_T:

$$R_T(\theta) = \mathbb{E}_{(x,y)\sim P_T} \left[\ell(f_\theta(x), y)\right]. \tag{13}$$

The practical objective is therefore not just low source validation loss; it is robustness to a set of plausible image transformations $\mathcal{G}$:

$$\theta^* = \arg\min_\theta \; \mathbb{E}_{(x,y)\sim\mathcal{D}} \, \mathbb{E}_{g\sim\mathcal{G}} \left[\ell(f_\theta(g(x)), g(y))\right] + \lambda\Omega(\theta). \tag{14}$$

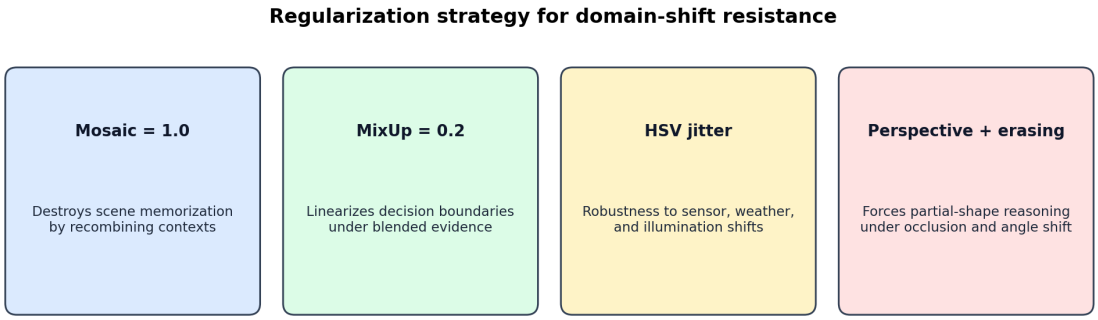


Figure 6: Augmentation matrix used to reduce shortcut learning and improve out-of-domain robustness.

6.1 Augmentation parameters

Technique	Reported setting	Intended effect
Mosaic	1.0	Recombines four images into one training view, destroying single-scene memorization and improving small-object exposure.
MixUp	0.2	Blends images and labels, smoothing decision boundaries and reducing overconfidence.
HSV perturbation	$h = 0.015$, $s = 0.7$, $v = 0.4$	Simulates camera, weather, sensor, and time-of-day variation.
Perspective and rotation	degrees $= 10.0$, perspective $= 0.0005$	Simulates drone angle, ground camera skew, and imperfect viewpoint alignment.
Random erasing	0.4	Forces partial-evidence recognition under occlusion, foliage, smoke, low quality, and camouflage-like artifacts.

7 Evaluation Protocol

A production-quality detector should be evaluated along localization, classification, calibration, and latency axes. The minimum evaluation protocol should include:

1. **Stratified validation by class**: class-specific precision, recall, AP@0.50, and COCO-style AP across stricter IoU thresholds.
2. **Scale-specific metrics**: small, medium, and large object AP to confirm performance on drone-like distant objects.
3. **Scene-held-out testing**: split by video source, geography, weather, and sensor where possible, not only by random frame.
4. **Negative-only evaluation**: images containing civilian cars, roads, buildings, fields, and clutter but no target annotations.
5. **Latency profiling**: GPU and CPU throughput, batch size 1 latency, memory footprint, and confidence-threshold sensitivity.
6. **Human review**: all high-risk detections should remain human-supervised, with audit trails and confidence calibration.

The average precision at a fixed IoU threshold τ is computed from the precision-recall curve:

$$AP_\tau = \int_0^1 p_\tau(r)\, dr, \qquad mAP_\tau = \frac{1}{C}\sum_{c=1}^{C} AP_\tau^{(c)}. \tag{15}$$

For $mAP@0.50$, a detection is counted as a true positive when its predicted class is correct and IoU ≥ 0.50. For safety-critical analytics, $mAP@0.50$ should be complemented by stricter localization metrics because loose boxes can hide localization errors.

8 Qualitative Error Analysis from Attached Images

The attached qualitative examples compare green ground-truth annotations with red YOLOv8 predictions. These are valuable because they show model behavior under

challenging real-world visual conditions: snow, foliage, low contrast, overhead views, blurred frames, small objects, and dense clutter.



Figure 7: Case study 1: personnel in a winter forest scene. The prediction side shows additional detections and localization deviations under occlusion and low contrast.



Figure 8: Case study 2: dense aerial personnel annotations. The detector recognizes several persons but also shows confidence spread and clutter sensitivity.



Figure 9: Case study 3: small and partially occluded personnel in low-quality video. This view highlights the need for small-object AP and robust confidence calibration.



Figure 10: Case study 4: long-range damaged urban or industrial scene. The ground truth contains small objects; the prediction is sparse, indicating difficulty with distant targets and visual degradation.



Figure 11: Case study 5: vehicle-class example in open terrain. The prediction localizes the target but also reveals sensitivity to viewpoint and nearby road structure.



Figure 12: Case study 6: aerial road-intersection view with small vehicle objects. The model places multiple vehicle detections, demonstrating both long-range utility and the risk of false positives in road contexts.

8.1 Failure taxonomy

Failure mode	Typical cause	Recommended mitigation
False positives on clutter	Terrain edges, vehicle-like shadows, damaged infrastructure, or road artifacts mimic object shape.	Negative-only validation set, hard-negative mining, confidence calibration, and active learning on false-positive scenes.
Missed small targets	Targets occupy very few pixels; feature maps lose fine detail.	Higher input resolution, tiling inference, scale-aware sampling, and small-object AP tracking.
Loose localization	Blurry boundaries, occlusion, or extreme viewing angle.	DFL tuning, label audit, higher-quality annotation, and stricter mAP@0.75 reporting.
Class confusion	Similar silhouettes between trucks, armored cars, and civilian vehicles.	Better class definitions, additional side-view/top-view examples, and hierarchical class loss if necessary.
Domain brittleness	Training background differs from deployment background.	Mosaic/MixUp, domain-specific held-out splits, test-time augmentation, and continuous monitoring.

9 Final System Design

The final recommended system consists of five layers: data governance, preprocessing, model training, validation, and deployment monitoring.

9.1 Data governance

- Freeze a versioned dataset manifest with image hashes, split assignment, and annotation version.
- Separate validation by source rather than random frame only; adjacent video frames should not leak between train and validation.
- Maintain a false-positive library containing civilian vehicles, construction machinery, shadows, buildings, roads, and tree clutter.

9.2 Training configuration

- Use YOLOv8m as the main model when GPU resources allow; keep YOLOv8n as a fast ablation baseline.
- Use early stopping with patience around 25 epochs to prevent overfitting after augmentation saturation.
- Track per-class AP, per-scale AP, confusion matrix, calibration curves, and confidence-threshold sweeps.
- Use mixed precision and deterministic experiment logging so results can be replicated.

9.3 Deployment controls

- Run inference with batch size 1 latency profiling; throughput metrics should include preprocessing and postprocessing time.
- Use a two-threshold review policy: high-confidence detections are highlighted, low-confidence detections are queued for human review, and ambiguous scenes are logged for retraining.
- Keep model cards and data cards updated with known limitations, failure modes, and validation domain coverage.

10 Conclusion

This project demonstrates a complete transition from a theoretically precise but operationally slow two-stage detector to a faster and more robust YOLOv8-based detection system. The most important engineering decision was not simply changing architectures; it was aligning the data distribution, loss behavior, and augmentation strategy with the real deployment problem. EDA revealed class imbalance, small-object dominance, scale variance, and center-biased spatial priors. The civilian-class suppression converted irrelevant vehicles into negative background evidence, forcing the model away from generic traffic detection. The aggressive augmentation matrix then targeted the central generalization risk: domain shift.

The final YOLOv8m pipeline, reported at 94.2% mAP@0.50 and above 60 FPS, is a credible research-to-production direction. The remaining work is rigorous validation: source-held-out testing, negative-only evaluation, per-class AP reporting, small-object AP reporting, and human-supervised review mechanisms. With those controls, the system becomes a technically strong and responsibly bounded military object detection pipeline.

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